INCH-POUND
MIL-DTL-23765C(SH)
16 August 2024
SUPERSEDING
MIL-E-23765B(SH)
22 August 1985

DETAIL SPECIFICATION

ELECTRODES AND RODS – WELDING, BARE, SOLID AND ALLOY-CORED, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea System Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bare, solid and alloy-cored welding electrodes and cut-length rods made from carbon steel, low alloy steel, and copper-base alloys designed for use with the gas metal-arc welding (GMAW), gas tungsten-arc welding (GTAW), and the submerged-arc welding (SAW) processes, as applicable.

1.2 Classification. Electrodes and rods are furnished in the forms and sizes, specified in 1.2.1 and 1.2.2, and in the type specified in the applicable specification sheet (see 6.2). Alloy-cored electrodes (see 6.8.1) for SAW are designated by a "C" suffix to the MIL-type designation for the bare, solid electrodes in the applicable specification sheet.

1.2.1 Forms. Electrodes and rods are furnished in the following forms:

a. Form 3a - Electrode on small spool (4-inch flange)
b. Form 3b - Electrode on large spool (8- and 12-inch flange)
c. Form 3c - Electrode wound on coils with support
d. Form 3d - Electrode wound in coils (65 pounds and under)
e. Form 3e - Electrode wound in coils (over 65 pounds)
f. Form 4 - Electrode wound in drums
g. Form 6 - Cut-length rods in containers

1.2.2 Sizes. Electrodes and rods are furnished in the sizes specified in table I.

TABLE I. Diameter sizes.

Form	Diameter sizes (inch)
3a	0.020, 0.025, 0.030, 0.035, and 0.045
3b	0.020, 0.025, 0.030, 0.035, 0.045, 0.052, 1/16, 5/64, 3/32, 1/8, 5/32, 3/16, 7/32, and 1/4
3c, 3d, 3e, and 4	0.030, 0.035, 0.045, 0.052, 1/16, 5/64, 3/32, 1/8, 5/32, 3/16, 7/32, and 1/4
6	0.030, 0.035, 0.045, 0.052, 1/16, 5/64, 3/32, 1/8, 5/32, 3/16, 7/32, and 1/4

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-10430	-	Welding Rods and Electrodes; Packaging of
MIL-DTL-23765/1	-	Electrodes and Rods – Welding, Bare, Solid and Alloy-Cored, Ordinary Strength and Low-Alloy Steel
MIL-E-23765/3	-	Electrodes and Rods – Welding, Bare, Solid Copper Alloy
MIL-DTL-23765/4	-	Electrodes – Welding, Base, Solid; and Fluxes, Submerged-Arc Welding, Carbon and Low-Alloy Steels
MIL-DTL-23765/5	-	Electrodes and Rods – Welding, Bare, Solid, High Yield Strength Low Alloy Steel, Stress-Relieved Weld Applications

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A5.01M/A5.01 - Welding and Brazing Consumables - Procurement of Filler Metals and Fluxes

AWS A5.02/A5.02M - Specification for Filler Metal Standard Sizes Packaging and Physical Attributes

AWS B4.0 - Standard Methods for Mechanical Testing of Welds

(Copies of these documents are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM D635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications.

ASTM E350 - Standard Test Methods for Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron, and Wrought Iron

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60974 - Arc Welding Equipment

(Copies of this document are available online at www.iec.ch.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org).

National Electrical Manufacturers Association (NEMA)

NEMA EW 1 - Electric Arc-Welding Power Sources

(Copies of this document are available online at www.nema.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. When specified in the specification sheet (see 3.1 and 6.2), electrodes and rods shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.3).

3.3 First article. When specified (see 3.1 and 6.2), a sample of one set of copper alloy electrodes and cut-length rods in accordance with MIL-E-23765/3 shall be subjected to first article inspection (see 4.3 and 6.4).

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3.4 Material.

3.4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.2 Asbestos. The electrodes, rods, fluxes, and their containers shall be asbestos-free (see 3.9.2 and 4.4.8.9).

3.4.3 Material from previous production runs. Materials from previous production runs shall be excluded from any new production run to be offered for acceptance, unless the materials have been determined to be acceptable for meeting all requirements of this specification (see 4.6.2.1).

3.5 General requirements.

3.5.1 Finish. Electrodes and rods shall have a smooth finish that is free from slivers, depressions, scratches, seams, laps (exclusive of the longitudinal joint in alloy-cored electrodes), scale, and foreign matter that would adversely affect the welding characteristics, operation of the equipment, or properties of the deposited metal (see 4.6.2.2).

3.5.2 Chemical composition. The chemical composition of bare, solid electrodes (wire form) and rods or weld metal deposited by alloy-cored electrodes shall be as specified in the specification sheet (see 4.6.2.3 and 6.3.3).

3.5.3 Mechanical properties. The deposited weld metal shall exhibit mechanical properties specified in the specification sheet (see 4.6.2.4).

3.5.4 Soundness. Radiographs of groove welds shall show that the welds conform to class 1 production weld requirements in accordance with MIL-STD-2035 (see 4.6.2.5).

3.5.5 Alloy identity.

3.5.5.1 All phases of production prior to final marking. The material shall be of the type specified when tested in accordance with 4.6.2.6 during all phases of production prior to final marking.

3.5.5.2 After final marking. When specified (see 6.2), each electrode and rod shall be of the material type specified when tested in accordance with 4.6.2.6 after final marking. Identification of each spool, coil, or container of electrodes tested after final marking and meeting the acceptance criteria shall include the words “alloy tested”.

3.6 Spooled and coiled electrodes.

3.6.1 Tolerance (diameter size). Diameter of electrode shall not vary from the nominal diameter by more than the limits identified in AWS A5.02/A5.02M (see 4.6.2.7).

3.6.2 Winding. Each spool, coil, and drum shall be wound of one continuous length of electrode made from a single lot of metal, as specified in the specification sheet. The winding shall avoid producing kinks, waves, or sharp bends and shall be free to unwind without restriction. The starting end of spooled and coiled electrode shall be firmly fastened so that it is readily visible to the welder or welding operator, and both ends of the electrode shall be accessible for inspection purposes (see 4.6.2.8).

3.6.3 Spool and coil weights. Spools and coils shall be furnished in the weights specified in the applicable specification sheet (see 4.6.2.9).

3.6.4 Cast and helix. The cast and helix of the unrestrained circle formed using the procedure in 4.6.2.10 shall be in accordance with table II.

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TABLE II. Cast and helix requirements.

Electrode form	Bare, solid electrode			Alloy-cored electrode	
	Cast (inches)		Helix (inches) 1/ (maximum)	Cast	Helix
	(minimum)	(maximum)			
3a	6	9	½	4/	4/
3b (0.030 and smaller)	12	30	----		
3b (0.035 and larger)	15	36	1½		
3c, 3d, 3e, and 4	2/	3/	2		

NOTES:

1/ Vertical separation between overlapped ends.

2/ Not less than the original inside coil diameter.

3/ Not greater than 2.5 times the original inside coil diameter.

4/ Alloy-cored electrodes shall feed uniformly and uninterrupted on automatic and semiautomatic welding equipment.

3.6.5 Spool and coil dimensions. (See 4.6.2.11)

3.6.5.1 Forms 3a and 3b. Spools used for form 3a and 3b electrodes shall be in accordance with AWS A5.02/A5.02M for the equivalent flange diameter.

3.6.5.2 Forms 3c, 3d, and 3e. Coil dimensions shall be as specified in table III.

TABLE III. Coil dimensions and accessories.

Form	Width (maximum) (inches)	Outside diameter (maximum) (inches)	Inside diameter (inches)
3c	3	17	12±¼
3d	4⅝	17	12±¼
3e	5	31½	22½ minimum

3.6.6 Identification. (See 4.6.2.12)

3.6.6.1 Forms 3a and 3b. The specification number, MIL-type, size, date of processing, and lot control number shall appear on the flange of the spool so as to be visible when the spool is positioned on the machine for welding. Such identification shall be placed in a manner that is not readily removable.

3.6.6.2 Forms 3c, 3d, 3e, and 4. Tags made of material that will resist effacement or destruction and bearing the specification number, MIL-type, size, date of processing, and lot control number shall be firmly attached at the inner end of the coiled filler metal. Coils wound on supports shall be identified with the above information placed directly on the support member in such a manner as not to be readily removable. Filler metal wound in drums shall be identified with the above information placed directly on the drum in such a manner as not to be readily removable.

3.6.7 Plastic spool burn characteristics. When specified (see 6.2), plastic spools shall not burn or shall stop burning when tested in accordance with 4.6.2.13.

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3.7 Form 6 (cut-length rods).

3.7.1 Tolerance (diameter size). Diameter of rods shall not vary from the nominal diameter by more than the limits identified in AWS A5.02/A5.02M.

3.7.2 Length. The length of rods shall be 12, 18, or 36 inches plus 0, minus ½ inch except for the 36-inch length rods, up to 10 percent of rods in any container may be shorter than 36 inches, but not shorter than 24 inches. Unless otherwise specified (see 6.2), the 36-inch length shall be furnished.

3.7.3 Identification. Cut-length rods shall be identified by positive and legible methods such as imprinting or indenting the applicable type designation number (see specification sheet) at one or more locations on the rod surface approximately 1 inch from the rod end, or shall be identified by pressure-sensitive, plastic-coated tape imprinted with applicable type number at one or more locations and attached to the rod approximately 1 inch from its end. Imprints on rods or on the tape shall be with fade-proof ink and shall be resistant to oils, solvents, all atmospheric conditions, and normal wear and tear encountered in shipping and handling.

3.8 Container requirements. Unit packs (containers) of electrodes, rods, and flux shall be in accordance with MIL-W-10430 preservation to level A, C, or commercial, or AWS A5.02/A5.02M, as specified (see 6.2). Additional requirements shall be as specified (see 4.6.2.14 and 6.2).

3.9 Container marking requirements. Markings for electrodes, rods, and flux shall be in accordance with MIL-W-10430 (see 4.6.2.14). Additional requirements shall be as specified (see 6.2).

3.9.1 Coils with liners, spools, or drums with and without support. In addition to special marking in accordance with MIL-W-10430 on warning labels, each coil with a liner, spool, or drum shall have the MIL-W-10430 specified warning label, or equivalent, permanently affixed to the body of the coil, spool, or drum in a prominent position and in legible type. Coils without support shall have a tag bearing the MIL-W-10430 specified warning label, or equivalent, in legible type securely attached to both ends of the coil.

3.9.2 Asbestos-free. Each unit container shall be marked “Asbestos-Free” (see 4.6.2.14).

3.10 Lot identification. Lot identification shall be accomplished by marking each unit and shipping container with a unique manufacturer’s (see 6.8.4) control number that is traceable to the original manufacturer’s records. If these control numbers are encoded, the interpretive key shall be furnished to the consignee. The control number shall be in a position that can be easily located by the purchaser (see 4.6.2.14). This control number shall be in a position that permits ready location by the consignee.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (see table IV):

a. Qualification inspection (see 4.2).

b. First article inspection (see 4.3).

c. Conformance inspection (see 4.4).

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TABLE IV. Qualification and conformance inspections.

Requirement description	Requirement paragraph	Verification paragraph	Verification type [1/]	Qualification	Conformance	First article
Asbestos	3.4.2	4.4.8.9	Certificate (when specified) Verification statement.	----	X	When specified in the specification sheet.
Material from previous production runs	3.4.3	4.6.2.1	VE	----	X	
Finish	3.5.1	4.6.2.2	VE or other	X	X	
Chemical composition	3.5.2 [2/]	4.6.2.3	Test	X	X	
Mechanical properties	3.5.3 [2/]	4.6.2.4	Test	X	X	
Soundness	3.5.4	4.6.2.5	Test	X	X	
Alloy identity	3.5.5	4.6.2.6	Test	When specified in the specification sheet.	When specified in the specification sheet.	
Spooled and coiled electrode diameter	3.6.1	4.6.2.7	DE	X	X	
Spooled and coiled electrode winding	3.6.2	4.6.2.8	VE	----	X	
Spool and coil weights	3.6.3	4.6.2.9	[3/]	----	X	
Cast and helix	3.6.4	4.6.2.10	DE	X	X	
Spool and coil dimensions	3.6.5	4.6.2.11	DE	----	X	
Spool and coil identification	3.6.6	4.6.2.12	VE	----	X	
Plastic spool burn test	3.6.7	4.6.2.13	Test	----	X	
Cut length rod diameter	3.7.1	4.6.2.7	DE	X	X	
Cut length rod length	3.7.2	4.6.2.7	DE	X	X	
Cut length rod identification	3.7.3	4.6.2.12	VE	----	X	
Containers	3.8	4.6.2.14	VE	----	X	
Container marking	3.9	4.6.2.14	VE	----	X	
Asbestos-free	3.9.2	4.6.2.14	VE	----	X	
Lot identification	3.10	4.6.2.14	VE	----	X	

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TABLE IV. Qualification and conformance inspections – Continued.

NOTES:

1/ The designations DE and VE are dimensional examination and visual examination, respectively.

2/ As specified in the specification sheet.

3/ Verification by weight.

4.2 Qualification inspection. Qualification inspection, when required in the specification sheet (see 3.1), shall include the examinations and tests specified in 4.5 (see 6.3).

4.2.1 Samples for qualification tests.

4.2.1.1 Electrode samples. Unless otherwise specified in the specification sheet, the electrode selected for testing shall be one spool of form 3b or one coil of form 3d, 1/16- or 1/8-inch diameter size of each MIL-type for which the manufacturer desires to qualify, provided the form and size is representative of the cleaning method used for all forms and sizes of that alloy. The 1/16-inch size shall be used for the GMAW process and the 1/8-inch size shall be used for the SAW process. If more than one cleaning method is used for various other diameter forms and sizes of any electrode alloy, qualification applications shall give complete details. The samples shall be selected in the presence of the Government representative (see 6.3.2).

4.2.1.1.1 Bare, solid electrode sample. Approval obtained for the 1/16-inch size electrode qualifies all sizes smaller than 3/32 inch of the same MIL-type. Approval obtained for the 1/8-inch size electrode qualifies all sizes 3/32 inch and larger for the same MIL-type.

4.2.1.1.2 Alloy-cored electrode sample. Alloy-cored electrodes shall be used with the SAW process. Approval obtained for the 1/8-inch size electrode qualifies all sizes 3/32 inch and larger for the same MIL-type that are made to the same formulation.

4.2.1.2 SAW flux sample. Unless otherwise specified in the specification sheet, the flux selected for testing shall be a standard size package of flux. The package shall accommodate the preparation of all required test assemblies. The package of flux shall remain unopened until the start of welding.

4.2.2 Change approval. Once a formulation is developed and qualified, and the aim values (composition validated by qualification testing) for each major alloying element have been established, changes to the formulation, aim values, construction, formulation, materials, design, or manufacturing process of qualified products may require requalification (see 6.3.3). This information is to be reported to NAVSEA, who will determine if requalification is required in cases where the changes are found to adversely affect design criteria or performance.

4.3 First article inspection. First article inspection shall consist of the examinations and tests in accordance with MIL-E-23765/3 (see 3.3).

4.3.1 First article inspection report. When first article inspection is required, a first article inspection report shall be prepared as specified (see 6.2 and 6.2.1).

4.3.2 Sample for first article inspection. The electrode selected for testing shall be one spool of form 3b or one coil of form 3d, 1/16-inch diameter size of each MIL-type for which the manufacturer desires to qualify, provided this form and size is representative of the cleaning method used for all forms and sizes of that alloy. If more than one cleaning method is used for various other diameter forms and sizes of any electrode alloy, first article applications shall give complete details. The samples shall be selected in the presence of the Government representative.

4.4 Conformance inspection. Conformance inspection shall include the examinations and tests specified in 4.5.

4.4.1 Lot definition of bare, solid electrodes.

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4.4.1.1 Lot. Unless otherwise specified in the specification sheet, for the purpose of sampling and inspection, a lot of electrodes or rods shall be the quantity of one type of alloy produced as specified in either 4.4.1.1.1 or 4.4.1.1.2.

4.4.1.1.1 Lot identification by heat of metal. Where manufacturing processes are such that wire is identified by a heat of metal, in addition to 4.4.1.1, a lot of electrodes shall be the quantity produced from a single heat of metal.

4.4.1.1.2 Lot identification by controlled wire chemical composition. Where manufacturing processes are such that wire is identified by controlled chemical composition of each MIL-type, in addition to 4.4.1.1, a lot of electrodes shall be the quantity produced from one or more chemically tested mill coils from one or more heats of metal conforming to the approved chemistry control limits for that type electrode. The following additional conditions shall apply:

a. Each continuously rolled mill-coiled rod furnished by mill prohibiting spliced-coil practice shall be chemically analyzed by approved methods. Mill-coiled rod furnished by mills permitting spliced-coil practice shall have no more than one splice per coil, and both ends of every coil received from such mills shall be chemically analyzed before processing.

b. Chemical analyses shall be certified by the laboratory and made available to the Government representative.

c. Mill coils conforming to established wire chemistry control for a specific MIL-type electrode shall be appropriately identified and segregated to avoid mix-ups.

4.4.2 Lot definition of alloy-cored electrodes.

4.4.2.1 Lot. A lot of electrodes shall be the quantity of any size and type produced from the following:

a. One dry batch (see 4.4.2.2) or one dry blend (see 4.4.2.3) and one heat of metal (see 4.4.2.4) not to exceed 25,000 pounds of electrodes, or

b. Chemically controlled dry batches of flux material (see 4.4.2.6) and chemically controlled mill coils (see 4.4.2.7) not to exceed 100,000 pounds of electrodes.

4.4.2.2 Dry batch. A dry batch of alloy mixture shall be the quantity of dry alloy ingredients mixed at one time in one mixing vessel. The dry batch may be used singly or subsequently subdivided.

4.4.2.3 Dry blend. A dry blend shall be one or more dry batches mixed in a mixing vessel and combined proportionately to produce a uniformity of mixed ingredients equal to that obtained by mixing the same total amount of dry ingredients at one time in one mixing vessel.

4.4.2.4 Heat of metal. A heat of metal shall be the material obtained from one furnace melt.

4.4.2.5 Chemically controlled materials. The procedure for manufacturing alloy-cored electrodes using chemically controlled material shall be as specified in 4.4.2.6 for flux material and 4.4.2.7 for mill coil.

4.4.2.6 Flux material. Each dry batch (see 4.4.2.2) shall be chemically analyzed for conformance to the percent allowable variation (disclosed) from standard (not disclosed) for each formulated chemical element for each MIL-type electrode. The following conditions shall apply:

a. Identification of each dry batch as to the intended use based on the results of the chemical analysis.

b. Maintain a record of chemical analysis.

c. Chemical analysis report of allowable variation from standard shall be available to the Government representative upon request.

d. Maintain traceability to each dry batch in lot identification of the electrodes.

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4.4.2.7 Mill coil. Each continuously rolled mill coil shall be chemically analyzed for conformance to the approved chemical composition control limits (see 6.3.1.4). The mill coils used for the production of one lot of electrodes shall be the product of one or more heats of metal. The following conditions shall apply:

a. One end of each mill coil furnished by mills prohibiting spliced-coil practice shall be sampled for the above chemical analysis. Both ends of each mill coil furnished by mills permitting one splice per coil shall be sampled for the above chemical analysis. Mill coils with more than one splice in each coil shall not be used.

b. Chemical analysis shall be certified by the laboratory and made available to the Government representative.

c. Mill coils that conform to the established chemical composition control limits for a specific MIL-type electrode shall be appropriately identified and segregated to avoid mix-ups.

4.4.3 Lot definition of SAW flux.

4.4.3.1 Inspection lot. For purposes of sampling and conformance inspection, a lot shall consist of all flux of one type offered for delivery at one time.

4.4.3.2 Lot identification. Each lot of flux shall be uniquely identified by the manufacturer's control number or other marking, which shall appear on each unit and shipping container.

4.4.4 Sampling for rods and electrodes.

4.4.4.1 Sampling for inspection of filled containers. Unit packages and shipping containers shall be sampled and inspected in accordance with MIL-W-10430.

4.4.4.2 Sampling for visual examination of rods and electrodes. Sample rods and electrodes from each lot shall be selected either from the production line immediately prior to packaging or from filled unit packages. If selected from the production line, the total sample shall be in accordance with table V and the electrodes or rods shall be selected throughout the run so that all parts of the run are represented. If the selection is made after the packaging operation, the total sample shall be in accordance with table V and approximately the same number of electrodes or rods shall be selected from each of the sample unit packages (five cut-lengths [rods] of each package) and an electrode length that will form one complete circle shall be taken from each sample spool, coil, and drum.

TABLE V. Sampling for visual examination and cast and helix tests.

Lot size, number of spools, coils, or drums of electrodes or boxes of rods in lot	Sample size, number of spools, coils, drums, or boxes selected at random for examination
2 to 8	2
9 to 15	2
16 to 25	3
26 to 50	3
51 to 90	5
91 to 150	5
151 to 280	8
281 to 500	8
501 to 1,200	13
1,201 to 3,200	13
3,201 to 10,000	20
10,001 to 35,000	20

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Any sample coil, spool, drum, or box that contains any nonconforming material shall be counted as defective. The maximum allowable number of defects shall be as specified (see 6.2 and 6.9). If in any sample the number of coils, spools, drums, or boxes that do not conform to this specification exceeds the acceptance number specified (see 6.2), this shall be cause for rejection of the lot.

4.4.4.3 Sampling for cast and helix tests. Specimens shall be cut from spools and coils selected in accordance with 4.4.4.2 and table V and shall be tested and evaluated as specified in 3.6.4. If any specimen fails the tests, the tests shall be performed on every spool or coil of the lot (see 4.4.1.1 as applicable) and all failures shall be rejected.

4.4.4.4 Sampling for chemical analysis. One specimen shall be selected from each lot (see 4.4.1 or 4.4.2) and chemically analyzed to verify conformance to the required chemical composition specified in the specification sheet. When more than one production line is being used to deposit copper coating on a single lot of electrodes, the following procedure shall be conducted in addition to the usual chemical composition conformance test.

a. Sample the product of each line at least once each shift.

b. Analyze for copper coating and record each result.

c. Identify the product of each line.

d. Purge nonconforming material prior to shipment.

4.4.4.5 Sampling for weld tests. A quantity of electrodes or rods shall be selected from each lot (see 4.4.1.1) to accommodate all weld tests if required herein and in the specification sheet.

4.4.5 Sampling for SAW flux.

4.4.5.1 Sampling for examination of filled containers. A random sample of filled containers from each lot shall be selected, examined, and accepted from each lot in accordance with MIL-W-10430. The maximum allowable number of defects shall be as specified (see 6.2 and 6.9).

4.4.5.2 Sampling for weld test. A quantity of flux shall be selected from each lot (see 4.4.3) to accommodate all weld tests required herein and in the specification sheet.

4.4.6 Sampling for container, container marking, and lot identification examinations. Unless otherwise specified (see 6.2), sampling of each lot for verification of conformance to container, container marking, and lot identification requirements of section 3 shall be in accordance with MIL-W-10430 (see 3.8, 3.9, and 3.10).

4.4.7 Sampling for plastic spool burn test. Sampling shall be as specified (see 3.6.7 and 6.2).

4.4.8 Certification of conformance. Unless otherwise specified (see 6.2), a certificate of conformance (CoC) shall be prepared for each lot of material offered for acceptance in accordance with the lot definition as specified in the specification sheet. The CoC shall be in accordance with the requirements herein and the specification sheet. A CoC or equivalent test certificate (see 4.4.8.8) shall be furnished to the consignee with the shipment and shall meet the following requirements.

4.4.8.1 Test results. The certificate shall include quantitative results for specified tests herein and the specification sheet, as applicable, for chemical, mechanical, and diffusible hydrogen tests. The certification shall also include qualitative results of nondestructive tests and other inspections and tests. The cause for any retest shall be reported for information and the test results that failed to meet those specified requirements shall be reported.

4.4.8.2 Stress relief. A record of any stress relief treatment (if applicable) shall also be reported.

4.4.8.3 Electrode flux combination. When required to use a specific electrode and flux combination, the certification shall report the brand name of flux used in the conformance testing.

4.4.8.4 Lot information. Except for controlled mill coil chemical composition lots (see 4.4.1.1.2), the heat number and material source (melter) shall be reported for lots identified by heat of metal (see 4.4.1.1.1).

4.4.8.5 Welding information. Welding process and operational parameters for welded test assemblies shall also be reported.

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4.4.8.6 Other information. The manufacturer, distributor customer, order number, specification, MIL-type, heat and lot, size, and form shall be identified.

4.4.8.7 Certification statement. The certificate shall state that each lot has been sampled, tested, and inspected in accordance with the specification requirements to produce an electrode, rod, or flux that meets the requirements. The certificate shall be signed by a responsible representative of the manufacturer.

4.4.8.8 Test certificates. Where test certificates issued by the manufacturer contain the above information (see 4.4.8 thru 4.4.8.7), a separate CoC will not be required.

4.4.8.9 Asbestos-free. A CoC indicating electrodes, rods, flux, and packaging are asbestos-free shall be submitted to the contracting activity when specified (see 3.4.2 and 6.2).

4.5 Examinations and tests. Samples selected in accordance with 4.2 and 4.4, as applicable, shall be subject to the examinations and tests for qualification, conformance, and first article inspections as designated in table IV and as specified in the specification sheet. If the results of any conformance test or examination do not satisfy the applicable requirements, then the lot shall be rejected, except as permitted in 4.4.4.3.

4.5.1 Procedures and methods. Procedures and methods shall be as specified in 4.6.

4.5.2 Unsatisfactory test results. Retest, when performed, shall be in accordance with 4.6.4.

4.6 Procedures and methods.

4.6.1 Test weld fabrication.

4.6.1.1 Welding machines. Manual, semi-automatic, or machine equipment shall be used with either the GMAW or GTAW processes for testing electrode or cut-length rod. The equipment used for SAW shall be a standard commercial type with automatic welding head, designed for use with and capable of a uniform and controlled feed of granular flux and bare wire, a uniform and controlled speed of travel, and of power capacity for the welding operations required in the specification sheet. For both qualification and conformance testing of electrodes and fluxes, welding machines used to supply power shall be of the following types: variable voltage direct current (DC) motor-generator type, variable voltage or constant voltage DC rectifier welder, variable voltage or constant voltage DC inverter, alternating current (AC) welding transformer, or an AC inverter, all of which shall be in accordance with NEMA EW 1 or IEC 60974. Multiple process machines are acceptable provided that they are properly set up. The power source shall be appropriate for the process to be tested and shall be of sufficient rating to supply current demanded by the electrode type and size under test. Open circuit voltage shall not exceed 80 volts. For welding using the pulsed gas metal-arc process, the model power supply shall be recorded. If applicable, the mode or program used for welding shall be recorded.

4.6.1.2 Instrumentation. For SAW and GMAW, voltage shall be measured between the wire feeder drive rolls and the work piece. For GTAW, voltage shall be measured between the test plate and a point as near as possible to the torch or the torch cable. Welding current and arc voltage measurements shall be made using digital or analog meters. Digital meters may be root mean square (RMS)-type devices provided the digital RMS meters are only used for direct current and non-pulsed arc welding; RMS meters shall not be used where alternating current or pulsed arc welding is performed. Tong meters shall not be used. Only calibrated instruments shall be used. Instrument calibration records shall be traceable to calibration performed in accordance with ISO/IEC 17025 or equivalent standard.

4.6.1.3 Shielding gas. Shielding gas shall be as specified in the specification sheet.

4.6.1.4 Welding parameters. Welding parameters used in all weld tests shall be as specified in the specification sheet.

4.6.1.5 Base plate. Base plate used in the preparation of welded joints shall be as specified in the specification sheet.

4.6.1.6 Flux. The flux used in depositing test welds with the SAW process shall be a neutral or active flux as specified in the specification sheet.

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4.6.2 Tests and examinations. Unless otherwise specified in the specification sheet, procedures and methods for tests and examinations for conformance to the section 3 requirements shall be in accordance with this paragraph. Test weld fabrication shall be in accordance with 4.6.1. Retesting, when performed, shall be in accordance with 4.6.4.

4.6.2.1 Pre-production inspection of equipment. Prior to the start of a new production run, where a different electrode type is to be produced, inspection shall be performed to assure that equipment and process lines are purged of material from the previous production run that do not meet the requirements herein, the requirements in the specification sheet, or conditions for qualification for the new production run. This inspection shall be performed for all equipment used in manufacturing operations where the material is not segregated and positively identified, for example cutting to length, extruding, and baking (see 3.4.3).

4.6.2.2 Finish. Examination for the finish shall be by visual examination, or other methods demonstrated to satisfy 3.5.1.

4.6.2.3 Chemical analysis. Chemical composition of the solid bare, solid electrode or rod (un-deposited) or weld metal deposited by alloy-cored electrodes shall be determined by any accepted analytical method agreed upon by the supplier and purchaser as specified (see 6.2). In case of dispute, the procedure in ASTM E350 shall be the referee method (see 3.5.2).

4.6.2.3.1 Chemical analysis of bare, solid electrodes and rods. For bare, solid electrodes and rods, chemical analysis shall be made on unwelded electrodes.

4.6.2.3.2 Chemical analysis for alloy-cored electrodes. For alloy-cored electrodes, chemical analysis shall be made on weld metal deposited as specified in 4.6.2.3.3 using the same brand flux used for qualification.

4.6.2.3.3 Chemical analysis of the weld deposit. When chemical analysis of the weld deposit is required, the sample shall be taken from a weld pad (see 4.6.2.3.3.1), or from a low dilution area in the groove weld or all weld metal tensile specimen. In case of dispute, the weld pad shall be the referee method.

4.6.2.3.3.1 Weld pad specifications. The weld pad shall be deposited in even layers in the flat position and shall be built up to not less than four layers, with three passes per layer as shown on figure 1. The final weld surface shall be ground flat for the chemical analysis and not less than three full surface layers shall remain after grinding. The welding parameters for the weld pad shall be selected to minimize weld dilution.

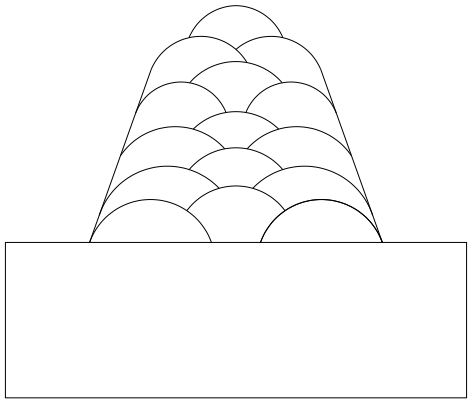


FIGURE 1. Chemical analysis weld pad schematic.

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4.6.2.4 Mechanical tests. When required in the specification sheet, tensile, Charpy V-notch (CVN), and bend test specimens shall be prepared and tested in accordance with AWS B4.0 (see 3.5.3).

4.6.2.4.1 Additional CVN specimen requirements. The CVN specimens shall be in the T-L orientation and centered in the weld joint to ensure that the notch is completely contained in weld metal.

4.6.2.4.2 Rounding-off procedures for tensile testing. Rounding-off procedures shall be as specified in 4.6.3.

4.6.2.4.3 Check mechanical property tolerances. Check mechanical property tolerances shall be as specified in 4.6.4.2.

4.6.2.5 Radiographic testing (RT) (soundness). Groove welds shall be radiographed in accordance with T9074-AS-GIB-010/271, using sensitivity level 2-2T (see 3.5.4).

4.6.2.6 Alloy identity. Alloy identity methods and procedures shall be in accordance with AWS A5.01M/A5.01 Department of Defense (DoD) supplementary requirements (see 3.5.5). The test method(s) and procedure(s) shall be submitted for approval to NAVSEA.

4.6.2.7 Electrode and rod dimensions (diameter and length). Diameter and length shall be verified (see 3.6.1 and 3.7).

4.6.2.8 Spooled and coiled electrode winding. Verification of spooled and coiled winding requirement shall be by visual examination (see 3.6.2).

4.6.2.9 Spool and coil weights. Spool and coil weight shall be verified (see 3.6.3).

4.6.2.10 Cast and helix. The cast and helix of spooled and coiled electrodes shall be measured by cutting a length required to impart to the electrode a curvature such that one loop is formed when placed on a flat surface forming an unrestrained circle for measurements in accordance with 3.6.4.

4.6.2.11 Spool and coil dimensions. Spool and coil dimensions shall be verified (see 3.6.5).

4.6.2.12 Spool, coil, and rod identification. Verification of spool, coil, and rod identification requirement shall be by visual examination (see 3.6.6 and 3.7.3).

4.6.2.13 Plastic spool burn tests. Plastic spool burn test procedures shall be in accordance with ASTM D635 (see 3.6.7).

4.6.2.14 Container, container marking, and lot identification examination. Container, container marking, and lot identification requirements shall be verified (see 3.8 and 3.9).

4.6.3 Rounding-off procedure. For the purposes of determining conformance with this specification, an observed or calculated value shall be rounded off to the nearest 1,000 pounds per square inch for tensile and yield strength and to the “nearest unit” in the last right-hand place of figures used in expressing the limiting value for other values in accordance with the rounding-off method specified in ASTM E29.

4.6.4 Unsatisfactory test results. Retest shall be in accordance with the following.

4.6.4.1 Retests. When any required original conformance test or examination representing a lot of material fails to meet specification requirements, the lot may be retested at the manufacturer’s discretion. Retest is not applicable to pre-production inspection of equipment (see 4.6.2.1) or container, container marking, and lot identification examinations (see 4.6.2.14). If the manufacturer chooses to retest the lot, the retests shall be as follows:

a. If the test failed due to one (or more) of the conditions listed in 4.6.4.1.1, then the lot shall be retested according to the requirements in 4.6.4.1.2.

b. If the test failed due to conditions other than those listed in 4.6.4.1.1, then the lot shall be retested according to the requirements in 4.6.4.1.3.

Note that all the specimens in a single test, whether the test involves a single specimen (such as for radiographic examination) or multiple specimens (such as for CVN impact tests), shall be considered a “set”.

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4.6.4.1.1 Testing or specimen preparation error retests. Retests to correct errors in testing and test specimen preparation are permitted under the following conditions:

a. Improperly fabricated test welds, such as test welds that have not been fabricated in accordance with the applicable specification or the weld procedure for the product under test.

b. Improperly prepared specimens, such as test specimen that are not in compliance with applicable standards.

c. Test equipment malfunction in the process of conducting a test.

d. Flaws in test specimens that are not indicative of inferior or defective product. Flaws at weld starts and stops at the ends of the test assembly, lack of fusion at the root, and slag due to operator technique are normal weld artifacts and do not count as flaws indicative of inferior or defective products. Cracks, clustered porosity, flakes, and ruptures are flaws that may be indicative of inferior or defective products and shall not qualify a lot for a testing or specimen preparation error retest.

4.6.4.1.2 Testing or specimen preparation error retest requirements. Retests for conditions listed in 4.6.4.1.1 shall be as follows:

a. For each original test that failed due to one (or more) of the conditions in 4.6.4.1.1, only one testing or preparation error retest is permitted.

b. A testing or preparation error retest shall consist of one replacement specimen for each specimen that failed due to the conditions in 4.6.4.1.1. Note for radiographic examination, the specimen is the entire weld. If the testing error is due to a malfunction of magnetic particle testing (MT) or RT equipment or to a flaw not indicative of inferior or defective products (see 4.6.4.1.1d) removed by grinding or other methods within $\frac{1}{16}$ inch of the surface, the original test assembly may be retested.

c. Results of all retests and reasons for all retests shall be reported on the certification of conformance.

d. Retests of the weld metal (e.g., mechanical properties) shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

e. Retests of the product (e.g., moisture, diffusible hydrogen, and chemistry pads) shall be from the same sampling as the original test, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

f. If the results of the testing or preparation error retest do not satisfy the applicable requirements, then a conformance retest shall be performed per 4.6.4.1.3.

4.6.4.1.3 Conformance retest requirements. Retests for failure to meet test requirements shall be as follows:

a. For each original test that failed due to conditions other than those listed in 4.6.4.1.1, only one conformance retest is permitted.

b. A conformance retest shall consist of repeating the test that failed two times (two-for-one). Note that a mechanical property test such as tensile or impact toughness involves a set of test specimens and, therefore, two full sets of specimens for each test that failed are required under this two-for-one rule. Results of all retests and reasons for all retests shall be reported on the certification of conformance.

c. Retests of weld metal shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

d. Retests of a product shall be from the same sampling as the original test, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

e. Unless the welding procedure was determined to be the root cause of the original test failure, any replacement test weld should employ the same welding procedure as the original test weld. If a new procedure is used to correct the root cause of the original test failure, the retest shall include all tests conducted for the original weld. Two-for-one is only required for the test that failed. All future lots of the material shall use the new procedure. The reason for changing procedure shall be justified to the customer and included on the certification.

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f. If the results of any conformance retest do not satisfy the applicable requirements, then the lot shall be rejected.

4.6.4.2 Check mechanical testing tolerances. Unless otherwise specified (see 6.2), retest as part of receipt inspection by a shipyard or other user of a product supplied in conformance with this specification shall use mechanical property tolerances to verify acceptability of the product as follows:

a. Yield strength test results (rounded off in accordance with 4.6.3) may be 3 percent higher than the specified maximum or 3 percent lower than the specified minimum requirements of the specification sheet.

b. Ultimate tensile strength test results (rounded off in accordance with 4.6.3), when required may be 3 percent lower than the specified minimum requirements of the specification sheet.

c. Elongation test results shall be not lower than 93 percent of the minimum requirement specified in the specification sheet.

d. CVN test results (both the average and the individual minimum value) may be lower than the requirements specified in the specification sheet by 10 percent.

4.7 Inspection of packaging and marking. Packaging and marking shall be inspected for compliance with the contract (see 5.1, 6.2, and 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified (see 6.2 and 6.5). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Bare and solid welding electrodes and cut-length rods are intended for use with the GMAW, GTAW, and SAW processes. Alloy cored welding electrodes are intended for use with SAW. This specification contains requirements above and beyond those of commercial specifications for use in critical military applications. Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet.

c. Types, forms, and sizes required (see 1.2).

d. When qualification is required (see 3.2).

e. When alloy identity testing of each electrode or rod after final marking is required (see 3.5.5.2).

f. When the plastic spool burn test is required and how many spools are to be tested (see 3.6.7 and 4.4.7).

g. When 12- or 18-inch rods are required (see 3.7.2).

h. Whether containers are required to be in accordance with MIL-W-10430 or AWS A5.02/A5.02M (see 3.8).

i. Additional container requirements (see 3.8).

j. Additional container marking requirements (see 3.9).

k. When first article inspection is required (see 4.3).

l. Maximum allowable number of defects (see 4.4.4.2 and 4.4.5.1).

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m. When a sampling plan for containers and container marking other than in accordance with MIL-W-10430 is required (see 4.4.6).

n. Plastic spool burn test sampling requirements (see 4.4.7).

o. When a CoC is not required (see 4.4.8).

p. When a CoC indicating electrodes, rods, flux, and packaging are asbestos-free is required (see 4.4.8.9).

q. Accepted analytical method agreed upon by the supplier and purchaser (see 4.6.2.3).

r. When check mechanical property tolerances are not permitted for retest as part of receipt inspection (see 4.6.4.2).

s. Packaging requirements (see 4.7, 5.1, and 6.5).

t. Requirements for first article inspection reports (see 6.2.1).

6.2.1 First article inspection report. The first article inspection report should be prepared as specified (see 4.3.1 and 6.2). Quantitative data for all chemical and mechanical tests, and qualitative data for all other tests and examinations, should be reported.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 23765 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Special instructions. When applying for test authorization, or after tests have been authorized and when samples are submitted, the following information will be requested. (This information, together with test results obtained with the sample, will form a part of the qualification test. All information will be held in confidence by the Government.)

6.3.1.1 Bare, solid electrode, heat of metal. For a bare, solid electrode, where lot identification is by heat of metal (see 4.4.1.1.1), the following information applies:

a. MIL-type and size under which approval is desired

b. Composition of the wire in terms of nominal percentages for each constituent

c. Composition of the deposited weld metal

d. Recommended amperages for each weld test

e. Brand name of the electrode and when applicable, brand name of the flux

6.3.1.2 Bare, solid electrode, controlled wire chemical composition. For a bare, solid electrode, where lot identification is by controlled wire chemical composition (see 4.4.1.1.2), in addition to 6.3.1.1, the following should be furnished for approval of the command or agency concerned (see 6.8.2):

a. Chemical composition control limits in the wire of each MIL-type electrode

b. Method of determining wire chemistry

c. Production line methods used to produce electrodes from chemically controlled core wire

6.3.1.3 Alloy-cored electrode, dry batch or blend filler and heat of metal. For an alloy-cored electrode, where lot identification is by a dry batch or dry blend of filler material and heat of metal (see 4.4.2), the following information applies:

a. Each type and size range (see 1.2) should be specified for which qualification is desired

b. Composition of tube, or strip, and flux ingredients in terms of nominal percentages for each constituent

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c. Composition of deposited weld metal

d. Recommended amperage for each weld test and size of electrode to be qualification tested

e. Brand name

6.3.1.4 Alloy-cored electrode, chemically controlled filler and mill coils. For an alloy-cored electrode, where lot identification is by chemically controlled dry batches of filler material and chemically controlled mill coils (see 4.4.2), the following information applies:

a. Each type and size range (see 1.2) should be specified for which qualification is desired

b. Chemical composition control limits for mill coil

c. Method of determining chemical composition of the mill coil

d. Production line quality control methods used in producing electrodes from chemically controlled mill coil

e. Percent of allowable variation (disclosed) from standard (not disclosed) for each formulated chemical element in the mix of chemically controlled filler material for each MIL-type electrode

f. Method of determining the chemical composition of the mix of chemically controlled filler material

g. Production line quality control methods used in producing electrodes from chemically controlled flux material

h. Recommended amperage for each weld test and the size of the electrode that is to be qualification tested

i. Brand name

6.3.1.5 SAW granular flux, manufacture control number. For SAW granular neutral or active flux, where lot identification is by the manufacture control number (see 4.4.3), the following information applies:

a. The type of flux for which qualification is desired

b. Percentage of allowable variation (disclosed) from standard (not disclosed) for each formulated chemical element in the chemically controlled flux for each MIL-type

c. Method of determining the chemical composition of the chemically controlled flux

d. Production line quality control methods used in producing the chemically controlled flux

e. Brand name

6.3.1.6 Aim values and typical ranges. The aim values and typical ranges for each element that represent expected variations due to manufacturing tolerances.

6.3.2 Samples for qualification inspection. Each manufacturer furnishes sample packages and electrodes of the type desired for inspection. Quantities are requested by the qualifying activity or laboratory and are selected at one time in the presence of the Government representative and so indicated. The manufacturer indicates on the sample whether the electrodes and packages were produced on a laboratory or experimental scale, or on a production scale.

6.3.3 Chemical analysis. The chemical compositions, as specified in the specification sheets, are minimum and maximum limits. However, the specification does not imply, and it should not be inferred, that each of the possible chemistry compositions defined by these limits will meet the full requirements of the specification, for example, mechanical properties.

6.4 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Packaging recommendations. The following minimum packaging recommendations are provided and are in addition to the requirements of section 3. Alternate or additional requirements may be necessary as determined by the purchaser.

6.5.1 Protection against damage. Electrodes, rods, and fluxes should be suitably packaged to ensure against damage during shipment and storage under normal conditions.

6.5.2 Packaging materials. Unless otherwise specified (see 6.2), packaging materials should be in accordance with MIL-W-10430.

6.5.2.1 Loose-fill packaging. The use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation or storage. If cushioning is required, the use of cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material should be specified.

6.5.2.2 Navy fire-retardant materials. When specified (see 6.2), Navy fire-retardant requirements for lumber, plywood, and fiberboard used in packaging should be in accordance with MIL-W-10430.

6.6 Inspection after delivery. Post-delivery inspection of electrodes, rods, flux, and packaging to determine conformance to this specification and for acceptance thereof, is the responsibility of the purchaser. Post-delivery inspection testing of electrodes intended for critical applications, such as submarine structures, should be in accordance with the applicable fabrication document.

6.7 Mixed-up material recommendations. If, subsequent to shipment, a heat or lot is found to contain material of a type or class different from that specified, the contractor (see 6.8.3) should notify all other customers who procured to the requirements herein and in the applicable specification sheet and received material from that heat or lot of the condition. The contractor should be held responsible for re-inspection to ensure that any unused material from the same heat or lot is correct, and should replace all incorrect material at no additional charge.

6.8 Definitions. For the purpose of this specification, the welding terms and definitions contained in AWS A3.0M/A3.0 must apply. In this specification and the specification sheets, the terms electrode and rod are used interchangeably with the term wire and denotes a filler-metal wire with no coating other than that incidental to its manufacturer or preservation, as specified in the specification sheet for coating.

6.8.1 Alloy-cored electrode. An alloy-cored electrode is a composite filler metal electrode consisting of a metal tube or other hollow configuration containing alloying ingredients. It has a formulation designed especially for the SAW process using a neutral granular flux.

6.8.2 Command or agency concerned. Command or agency Government, prime contractor, or authorized representative who has design or acquisition responsibility acting under contract to the Government.

6.8.3 Contractor. The contractor is the seller under the contract or purchase order which incorporates this specification.

6.8.4 Manufacturer. The manufacturer is the actual processor of the welding electrodes and rods engaged in the final cleaning, spooling, cutting to length, affixing rod identification, marking, and packaging operations.

6.9 Maximum allowable number of defects for visual examination samples. Table VI provides the maximum number of defects for samples selected in accordance with 4.4.4.2 and table V. These maximum numbers are the same as was provided in the previous revision of this specification.

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TABLE VI. Sampling for visual examination.

Lot size, number of spools, coils, or drums of electrodes or boxes of rods in lot	Sample size, number of spools, coils, drums, or boxes selected at random for examination	Acceptance number, maximum number spools, coils, drums, or boxes containing defective electrodes or rods for acceptance of the lot
2 to 8	2	0
9 to 15	2	0
16 to 25	3	0
26 to 50	3	0
51 to 90	5	0
91 to 150	5	0
151 to 280	8	1
281 to 500	8	1
501 to 1,200	13	2
1,201 to 3,200	13	2
3,201 to 10,000	20	3
10,001 to 35,000	20	3

6.10 Defective material replacement recommendations. When a lot is found to be acceptable when applying any fixed acceptable quality levels, lot tolerance percent defectives, or other types of fixed levels of defects the defective electrodes in the lot found to be defective should be replaced with non-defective electrodes prior to shipment. In no case should known defective material be shipped.

6.11 CoC considerations. The CoC as required in this specification is an essential element for assuring that the material being purchased is in conformance with the specification. The certificate content, as required herein, provides valuable information regarding the welding parameters used for the required tests, expected performance, actual weld metal chemistry, whether the product was subjected to a stress relief heat treatment, information regarding test failures, and other information that links the certificate to the manufacturer, purchaser, specification, lot, etc. Navy technical and safety program documentation typically require certain information included in the prescribed CoC for use as objective quality evidence demonstrating specification conformance and compliance as well as traceability for materials used within prescribed ship systems. Such documentation includes, but is not limited to, SUBSAFE; Level I; Deep Submergence Scope of Certification (DSSOC); and ship hull, machinery, and piping systems fabrication documents. A CoC may also be required by customs. The CoC also provides beneficial information in support of hull structure and component failure analyses; tracking problematic heats or lots of material throughout the supply system or onboard ships; and material, component, and specification improvements. These added benefits are possible because of the unique tracking information provided within the CoC and the unique set of mechanical properties and chemical content provided for each lot of material.

6.12 Subject term (key word) listing.

Gas metal-arc welding

Gas tungsten-arc welding

Submerged-arc welding

Welding electrodes

Welding rods

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6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 3439-2022-001)

Review activities:
Army – EA, MI
Navy – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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